

Online Resource 1

Supplementary material for “Quantum-like cognition and decision-making for autonomous agents: a survey from robotics to machine learning”, *Artificial Intelligence Review*. **[FILL: author names]**

This resource contains the classification of the 74 application studies by domain, model type, evidence level, task and main limitation (Table S1), and a glossary of symbols and notation (Table S2). Table S1 is also provided as a CSV file (Online Resource 3). Evidence levels: E1 conceptual; E2 formal model; E3 human data; E4 simulation or benchmark; E5 physical system; E6 field deployment. Citations refer to the reference list below.

Table S1 Application studies (n = 74): domain, model type, evidence level (Table 2 of the main article), task and main limitation

Study	Domain	Model type	Level	Task	Main limitation
Moreira and Wichert (2014)	Decision networks	Quantum-like	E2	Probabilistic inference in Bayesian networks under uncertainty	No human data; free interference phases
Moreira and Wichert (2016)	Decision networks	Quantum-like	E3	Sure-thing principle violations (disjunction effect)	Heuristic lacks strong psychological justification; aggregate data from few paradigms
Moreira (2017)	Decision networks	Quantum-like	E3	Modelling paradoxical human decisions with graphical models	Few paradigms; aggregate data; heuristic parameter setting
Moreira et al. (2020a)	Decision networks	Quantum-like	E3	Decision-making with utilities; paradoxical choices	Interference parameters still required; classic laboratory paradigms only
Wichert et al. (2020)	Decision networks	Quantum-like	E3	Predicting sure-thing principle violations	Two paradigms; aggregate data; maximum-entropy assumption
She et al. (2021)	Decision networks	Quantum-like	E4	Multi-attribute decision making with interacting attributes	Single application; cognitive interpretation of interference unclear
Meghdadi et al. (2022)	Decision networks	Quantum-like	E3	Biased human selection in decision games	Additional loosely constrained parameters; few aggregate datasets
Gao et al. (2025)	Decision networks	Quantum-like	E3	Sure-thing principle violations; information retrieval	Existing aggregate datasets; added model complexity
Widdows and Peters (2003)	ML, NLP and IR	Quantum-inspired	E4	Word-sense disambiguation / retrieval with logical connectives on word vectors	Qualitative evaluation only; no human data; non-distributivity not exploited empirically
Sordani et al. (2013)	ML, NLP and IR	Quantum-inspired	E4	Ad hoc document retrieval with term dependencies	Vocabulary-sized density matrices are costly; dependencies are hand-selected; no user/cognitive data
Lorenz et al. (2023)	ML, NLP and IR	Quantum circuit or hardware	E5	Sentence and noun-phrase binary classification (MC, RP)	Very small datasets, single hardware run, post-selection overhead, long queue times; no quantum advantage
Svozil (2025)	ML, NLP and IR	Quantum-inspired	E2	Contextual word embeddings / polysemy	No learning procedure, corpus or evaluation; practicality untested
Stoudenmire and Schwab (2016)	DL and LLMs	Quantum-inspired	E4	Image classification (supervised learning)	Single simple benchmark; 1-D ordering of 2-D inputs; no cognitive or agent evaluation
Zhang et al. (2018)	DL and LLMs	Quantum-inspired	E4	Question answering / answer selection (text matching)	Interpretability rests on analogy; text-matching only; no cognitive data
Li et al. (2019)	DL and LLMs	Quantum-inspired	E4	Question answering (semantic matching)	No gain over baselines; interpretability not user-tested
Li et al. (2021a)	DL and LLMs	Quantum-inspired	E4	Multimodal video sentiment analysis (fusion)	Task-specific; interpretability argued not measured; no cognitive data
Li et al. (2021b)	DL and LLMs	Quantum-inspired	E4	Emotion recognition in conversation (multimodal)	No gain over state of the art; benchmark-only; no human-judgement validation

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Gkoumas et al. (2021)	DL and LLMs	Quantum-like	E4	Multimodal decision fusion for video sentiment analysis	Incompatibility assumed not tested; benchmark labels, not human decision data
Upreti et al. (2024)	DL and LLMs	Quantum-like	E4	Order effects in similarity judgements by LLMs	Preliminary; prompt-dependent; no QP model fitted in abstract
Zhang et al. (2025)	DL and LLMs	Quantum-inspired	E4	Semantic matching / QA	Details unverified; density-matrix cost; interpretability not user-tested
Maksimovic and Maksymov (2025)	DL and LLMs	Quantum-inspired	E4	Image classification with human-like confidence/uncertainty	Human-likeness assessed via simulations, not participant data; datasets unverified
Lo et al. (2025)	DL and LLMs	Quantum-like	E4	Contextuality of LLM-derived probabilities in anaphora (co-reference) schema	Single masked LM; synthetic schema; no downstream-task benefit shown
Aerts et al. (2026)	DL and LLMs	Quantum-like	E4	Bell-inequality and quantum-statistics tests on LLM concept combination and text	Prompt/version dependence; signalling not controlled in abstract; interpretive leap to convergent cognition
Imannezhad et al. (2026)	DL and LLMs	Quantum-like	E4	Probability judgement fallacies (conjunction, disjunction, complementarity) in GPT-5 vs humans	Single proprietary LLM; personas as proxies; numbers to verify
Kang (2026)	DL and LLMs	Quantum-like	E4	Auditing question-order effects in LLM sequential judgements	Single small model; few items; saturated measurement interface
Dong et al. (2008)	Reinforcement learning	Quantum-inspired	E4	Reinforcement learning in unknown environments (gridworld navigation)	Only classical simulation of the quantum algorithm; small tabular tasks; claimed parallelism speed-up requires quantum hardware
Dong et al. (2012)	Reinforcement learning	Quantum-inspired	E5	Mobile-robot navigation and obstacle avoidance via RL	Tabular representation; single TD baseline; real-robot results qualitative
Briegel and De las Cuevas (2012)	Reinforcement learning	Quantum-inspired	E4	Learning agent design with internal simulation (projective simulation)	Toy environment only; classical simulation; no benchmark comparison with standard RL; quantum version only sketched
Paparo et al. (2014)	Reinforcement learning	Quantum circuit or hardware	E2	Speeding up deliberation (action selection) of active learning agents	No implementation or simulation of a full agent; requires coherent quantum memory; environment remains classical
Melnikov et al. (2018)	Reinforcement learning	Quantum-inspired	E4	Navigation benchmarks for RL agents (grid world, mountain car)	Small tabular tasks; classical only; no deep RL comparison
Chen et al. (2020)	Reinforcement learning	Quantum circuit or hardware	E4	Deep Q-learning for discrete decision tasks	Tiny environments; indirect classical comparison; no quantum advantage shown; hardware runs limited to few inference episodes
Jerbi et al. (2021)	Reinforcement learning	Quantum circuit or hardware	E4	Policy-gradient RL with parametrised quantum circuit policies	Separation relies on artificial tasks; no hardware runs; benchmark tasks are small
Wei et al. (2022)	Reinforcement learning	Quantum-inspired	E4	Experience replay sampling in deep reinforcement learning	Classical heuristic only; evaluation on game benchmarks; exact gains not verified here
Skolik et al. (2022)	Reinforcement learning	Quantum circuit or hardware	E4	Deep Q-learning with PQC function approximators	Simulation only; small benchmarks; PQCs unstable as parameters grow; separation limited to restricted environments
Acuto et al. (2022)	Reinforcement learning	Quantum circuit or hardware	E4	Continuous control of a simulated robotic arm (reaching)	Noise-free simulation only; simple planar arm; advantage limited to parameter count; long training times
Sinha et al. (2025)	Reinforcement learning	Quantum circuit or hardware	E4	Collision-free navigation of self-driving cars via deep RL	Simulated quantum circuits only; multi-day training; no clear driving-performance gain; noise degrades training
Hohenfeld et al. (2024)	Reinforcement learning	Quantum circuit or hardware	E4	Mobile-robot navigation via deep Q-learning	Noise-free simulation; circuits too deep for current hardware; quantum models lose in the realistic dynamic task
Lukac and Perkowski (2007)	Robots and embodied agents	Quantum circuit or hardware	E2	Generating emotional variation in humanoid robot behaviour	No evaluation or comparison with classical stochastic models; partial testing; simulated quantum layer

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Aerts et al. (2011b)	Robots and embodied agents	Quantum-like	E1	Conceptual bridge from quantum cognition to AI and robotics	No formal model for robots, no implementation or evaluation; speculative claims about computational power of the mind
Hangl et al. (2016)	Robots and embodied agents	Quantum-inspired	E5	Hierarchical manipulation skill learning by autonomous play	Small hand-designed state and skill sets; classical learning; limited task range
Yan et al. (2019)	Robots and embodied agents	Quantum circuit or hardware	E2	Representing and transforming robot emotional states	No empirical validation, implementation or classical comparison; benefits require quantum hardware
Moreira et al. (2020b)	Robots and embodied agents	Quantum-like	E1	Position on cognitive architectures for decision-making	No formal model specified in detail, no data or implementation; comparative claims not demonstrated
Lanza et al. (2020)	Robots and embodied agents	Quantum-like	E2	Robot perception of mutually exclusive percepts	Binary toy scenario; no classical baseline; hardware errors; no demonstrated perceptual benefit
Hangl et al. (2020)	Robots and embodied agents	Quantum-inspired	E5	Autonomous acquisition of robotic manipulation skills through play	Classical PS only; hand-designed skills; supervision needed; limited scalability evidence
Lanza et al. (2021)	Robots and embodied agents	Quantum-like	E2	Multisensory integration and belief representation in robot perception	Simulation only; no temporal integration; single query per prepared state; no classical fusion baseline
Yan et al. (2021b)	Robots and embodied agents	Quantum circuit or hardware	E2	Emotion generation and transition for companion robots	Apparently formal design only; no verified evaluation or classical comparison
Ho and Hoorn (2022)	Robots and embodied agents	Quantum circuit or hardware	E2	Modelling robot affect and user-directed attitude in HRI	No empirical validation or classical comparison; implementation status unclear
Chella et al. (2022)	Robots and embodied agents	Quantum circuit or hardware	E2	Robot path/motion planning on a grid map	Tiny maps; noiseless simulation only; static known environment; no runtime comparison with classical planners
Song et al. (2022)	Robots and embodied agents	Quantum-like	E4	Pedestrian crossing-intention estimation and trajectory prediction for autonomous vehicles	Narrow scenario; interference parameters not clearly learned; quantitative details not verified here
Widdows et al. (2023)	Robots and embodied agents	Quantum circuit or hardware	E5	Implementing quantum cognition models (order effects, disjunction/interference effects) as quantum circuits	No new human data or fitting; tiny circuits with no quantum advantage; hardware noise affects order-effect circuits
Mannone et al. (2024)	Robots and embodied agents	Quantum circuit or hardware	E2	Modelling robot 'thinking' and creative behaviour (improvised dance)	Toy model; hand-set states; no behavioural evaluation or classical baseline
Daglarli (2025)	Robots and embodied agents	Quantum circuit or hardware	E4	Autonomous behaviour of a smart agent (exploration, survival) in a game simulation	Quantum circuits classically simulated; largely qualitative reporting; unclear baselines and statistics; weak justification for algorithm choice
De Carolis et al. (2025)	Robots and embodied agents	Quantum circuit or hardware	E1	Review and project outline for quantum-enhanced social robotics	No implementation or evaluation; conflates quantum hardware, quantum-like and quantum-inspired work; limited critical appraisal
Chella et al. (2026)	Robots and embodied agents	Quantum circuit or hardware	E4	Reactive action selection from robot sensor data (wall following)	Offline only; no closed-loop robot control; no quantum advantage claimed; hand-defined rule partitions
Yukalov et al. (2018)	Multi-agent systems and teams	Quantum-like	E2	Multistep collective decision-making; dynamic disjunction effect	Stylised interaction rules; validated only against previously published aggregate data
Ried et al. (2019)	Multi-agent systems and teams	Quantum-inspired	E4	Collective motion / swarm alignment (marching locusts)	1-D toy world with two actions; qualitative match only; PS is classical here
Khoshnoud et al. (2020)	Multi-agent systems and teams	Quantum circuit or hardware	E5	Secure communication and coordinated action in cooperative aerial/ground multi-robot systems	Tutorial/concept demonstration; no key rates, error rates or latency measured; alignment acknowledged as coarse; speed claims do not address no-

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López-Incera et al. (2020)	Multi-agent systems and teams	Quantum-inspired	E4	Swarm foraging; emergence of collective motion	signalling 1-D world, two actions, simple reward; PS run classically
Lawless (2020)	Multi-agent systems and teams	Quantum-like	E1	Design and control of autonomous human-machine teams	Conceptual; analogies not formalised or tested quantitatively; case-study interpretation is post hoc
Yan et al. (2021a)	Multi-agent systems and teams	Quantum circuit or hardware	E2	Representing, transforming and fusing emotional states of social/multi-robot systems	Conceptual; simulation-only toy examples; no hardware execution, no empirical grounding of emotion model; exponential preparation cost and repeated measurement needed for read-out
Yun et al. (2022)	Multi-agent systems and teams	Quantum circuit or hardware	E4	Cooperative multi-agent reinforcement learning for edge-to-cloud packet offloading / queue management	Classical simulation only (no noise or hardware); single toy environment; advantage only under equal-parameter constraint; no repeated runs or variance reported
Yukalov et al. (2022)	Multi-agent systems and teams	Quantum-like	E2	Collective decision dynamics in agent networks	Idealised network and communication; no empirical calibration reported in abstract
Mannone et al. (2022)	Multi-agent systems and teams	Quantum circuit or hardware	E2	Modelling local-to-global interaction rules in robot swarms (two-robot target-reaching toy)	Toy two-robot model; no physical robots or classical baseline; efficiency claim not tested; high hardware noise; interpretation of amplitudes as decision uncertainty not validated
Yukalov (2023)	Multi-agent systems and teams	Quantum-like	E2	Affective (emotion-aware) decision-making for AI	No implementation or empirical test; architecture unspecified
Lawless et al. (2023)	Multi-agent systems and teams	Quantum-like	E1	Structure and performance of human-machine teams in open systems	Mainly review/position; analogical formalism; no new controlled data
Essalmi et al. (2025)	Multi-agent systems and teams	Quantum-inspired	E4	Interaction-aware decision-making for automated driving (merging, roundabouts)	Simulation only; 2×2 games; parameter-dependent; collision results reported as not reproducible by an independent review
Kong et al. (2026)	Multi-agent systems and teams	Quantum-inspired	E4	Group navigation and queue formation in swarms	Simulation only; PS classical; no physical robots
Ashtiani and Abdollahi Azgomi (2019)	Human-AI trust	Quantum-like	E4	Computational trust evolution in multi-agent/decision-support systems	Small synthetic scenarios; no human data; no quantum speed-up
Pushparaj et al. (2021)	Human-AI trust	Quantum-inspired	E3	Human-automation trust in air traffic control	Very small sample; no reported quantitative model comparison; limited generalisability
Humr et al. (2023)	Human-AI trust	Quantum-like	E2	Temporal evolution of human trust in AI-supported decisions	Short proceedings paper; validation not reported in abstract
Roeder et al. (2023)	Human-AI trust	Quantum-like	E3	Trust/reliability calibration in human-AI interaction	Exploratory; N = 7; scripted AI behaviour
Humr et al. (2025)	Human-AI trust	Quantum-like	E2	Human-in-the-loop decision-making with AI support	No new empirical data; proposals not implemented in an AI system
van der Meer et al. (2025)	Human-AI trust	Quantum-like	E3	Dynamics of human trust/reliability ratings in human-AI interaction	Preprint; quantitative evidence not given in abstract; small prior dataset
Humr and Canan (2026)	Human-AI trust	Quantum-like	E1	AI-supported (military) decision-making; human-AI alignment	QCAL not implemented or evaluated; GenAI probe small and qualitative

Table S2 Glossary of symbols and notation

Symbol	Meaning
$ \psi\rangle, \langle\psi $	State vector (ket) and its conjugate transpose (bra)
$\langle\phi \psi\rangle$	Inner product
ρ	Density matrix (mixed state); $\text{tr } \rho = 1, \rho \geq 0$

Symbol	Meaning
P_A	Projector onto the “yes” subspace of question A
$I - P_A$	Projector onto the “no” subspace
$[A, B] = AB - BA$	Commutator; zero for compatible questions
$\mathbb{P} \psi\rangle^2$	Born-rule probability
$U(t) = \exp(-iHt)$	Unitary evolution generated by Hamiltonian H
L_k, γ_k	Lindblad (jump) operators and rates in open-system models
$\otimes$	Tensor product of spaces or states
θ	Interference phase between two branches
QQ	Quantum question equality
QLBN	Quantum-like Bayesian network
QRL, QiRL	Quantum and quantum-inspired reinforcement learning
PQC, VQC	Parametrised and variational quantum circuit
PS	Projective simulation
CbD	Contextuality-by-default
POVM	Positive-operator-valued measure (generalised measurement)

References

- Acuto A, Barillà P, Bozzolo L, Conterno M, Pavese M, Policicchio A (2022) Variational Quantum Soft Actor-Critic for Robotic Arm Control. Preprint at arXiv:2212.11681
- Aerts D, Aerts Arguëlles J, Beltran L, Geriente S, Leporini R, Sassoli de Bianchi M, Sozzo S (2026) Identifying Quantum Structure in AI Language: Evidence for Evolutionary Convergence of Human and Artificial Cognition. *Entropy* 28(6):622. <https://doi.org/10.3390/e28060622>
- Aerts D, Czachor M, Sozzo S (2011b) Quantum Interaction Approach in Cognition, Artificial Intelligence and Robotics. In: *Proceedings of the Fifth International Conference on Quantum, Nano and Micro Technologies (ICQNM 2011)*. IARIA, pp 35–40. arXiv:1104.3345
- Ashtiani M, Abdollahi Azgomi M (2019) A novel trust evolution algorithm based on a quantum-like model of computational trust. *Cognition, Technology & Work* 21(2):201–224. <https://doi.org/10.1007/s10111-018-0496-9>
- Briegel HJ, De las Cuevas G (2012) Projective Simulation for Artificial Intelligence. *Scientific Reports* 2:400. <https://doi.org/10.1038/srep00400>
- Chella A, Gaglio S, Pilato G, Vella F, Zammuto S (2022) A Quantum Planner for Robot Motion. *Mathematics* 10(14):2475. <https://doi.org/10.3390/math10142475>
- Chella A, Gaglio S, Pilato G, Vella F (2026) An Architecture for a Quantum Teleo-Reactive Robot. *Entropy* 28(7):731. <https://doi.org/10.3390/e28070731>
- Chen SYC, Yang CHH, Qi J, Chen PY, Ma X, Goan HS (2020) Variational Quantum Circuits for Deep Reinforcement Learning. *IEEE Access* 8:141007–141024. <https://doi.org/10.1109/ACCESS.2020.3010470>
- Daglarli E (2025) A Generative Neuro-Cognitive Architecture Using Quantum Algorithms for the Autonomous Behavior of a Smart Agent in a Simulation Environment. *Computers, Materials & Continua* 84(3). <https://doi.org/10.32604/cmc.2025.065572>
- De Carolis BN, Loglisci C, Palestra G, Violante S (2025) Quantum-Enhanced Social Robotics: The QUADRI Project. In: *Adjunct Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP Adjunct '25)*. ACM, New York, NY, USA. <https://doi.org/10.1145/3708319.3735545>
- Dong D, Chen C, Li H, Tarn TJ (2008) Quantum Reinforcement Learning. *IEEE Transactions on Systems, Man, and Cybernetics, Part B: Cybernetics* 38(5):1207–1220. <https://doi.org/10.1109/TSMCB.2008.925743>
- Dong D, Chen C, Chu J, Tarn TJ (2012) Robust Quantum-Inspired Reinforcement Learning for Robot Navigation. *IEEE/ASME Transactions on Mechatronics* 17(1):86–97. <https://doi.org/10.1109/TMECH.2010.2090896>
- Essalmi K, Garrido F, Nashashibi F (2025) Quantum game models for interaction-aware decision-making in automated driving. Preprint at arXiv:2509.01582. arXiv preprint
- Gao X, Yang H, Pan L, Pelusi D (2025) Quantum-like evidence networks decision-making model. *Engineering Applications of Artificial Intelligence* 157:111368. <https://doi.org/10.1016/j.engappai.2025.111368>
- Gkoumas D, Li Q, Dehdashti S, Melucci M, Yu Y, Song D (2021) Quantum Cognitively Motivated Decision Fusion for Video Sentiment Analysis. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol 35, pp 827–835. <https://doi.org/10.1609/aaai.v35i1.16165>

- Hangl S, Dunjko V, Briegel HJ, Piater J (2020) Skill Learning by Autonomous Robotic Playing Using Active Learning and Exploratory Behavior Composition. *Frontiers in Robotics and AI* 7:42. <https://doi.org/10.3389/frobt.2020.00042>
- Hangl S, Ugur E, Szedmak S, Piater J (2016) Robotic Playing for Hierarchical Complex Skill Learning. In: *Proceedings of the 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp 2799–2804. arXiv:1603.00794
- Ho JKW, Hoorn JF (2022) Quantum Affective Processes for Multidimensional Decision-Making. *Scientific Reports* 12:20468. <https://doi.org/10.1038/s41598-022-22855-0>
- Hohenfeld H, Heimann D, Wiebe F, Kirchner F (2024) Quantum Deep Reinforcement Learning for Robot Navigation Tasks. *IEEE Access* 12:87217–87236. <https://doi.org/10.1109/ACCESS.2024.3417808>
- Humr S, Canan M (2026) System Level Intervention for AI-Supported Decision-Making: A Quantum Cognition Perspective. In: *Quantum AI and NLP (QNLPAI 2025)*. *Communications in Computer and Information Science*, vol 2659. Springer, Cham, pp 250–273. https://doi.org/10.1007/978-3-032-13883-5_14
- Humr SA, Canan M, Demir M (2023) Temporal Evolution of Trust in Artificial Intelligence-Supported Decision-Making. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 67(1):1936–1941. <https://doi.org/10.1177/21695067231193672>
- Humr S, Canan M, Demir M (2025) A Quantum Probability Approach to Improving Human–AI Decision Making. *Entropy* 27(2):152. <https://doi.org/10.3390/e27020152>
- Imannezhad P, Pothos EM, Wills AJ (2026) Divergent Patterns of Probabilistic Reasoning in Humans and GPT-5. *Frontiers in Psychology* 17:1782184. <https://doi.org/10.3389/fpsyg.2026.1782184>
- Jerbi S, Gyurik C, Marshall SC, Briegel HJ, Dunjko V (2021) Parametrized Quantum Policies for Reinforcement Learning. In: *Advances in Neural Information Processing Systems*. Vol 34. arXiv:2103.05577
- Kang P (2026) Auditing Question-Order Effects in Large Language Models with the QQ Equality: Mechanism Characterization and a Saturation Caveat. Preprint at arXiv:2607.17219
- Khoshnoud F, Quadrelli MB, Esat II, Robinson D (2020) Quantum Cooperative Robotics and Autonomy. Preprint at arXiv:2008.12230
- Kong D, Xue K, Wang P, Xu Z (2026) Emergence of Longitudinal Queues in Group Navigation: An Interpretable Approach via Projective Simulation. *Biomimetics* 11(3):201. <https://doi.org/10.3390/biomimetics11030201>
- Lanza D, Solinas P, Mastrogiovanni F (2020) A Preliminary Study for a Quantum-like Robot Perception Model. Preprint at arXiv:2006.02771
- Lanza D, Solinas P, Mastrogiovanni F (2021) Multi-sensory Integration in a Quantum-Like Robot Perception Model. In: Siciliano B, Laschi C, Khatib O (eds) *Experimental Robotics: The 17th International Symposium (ISER 2020)*. Springer Proceedings in Advanced Robotics, vol 19. Springer, Cham, pp 502–509. https://doi.org/10.1007/978-3-030-71151-1_44
- Lawless WF (2020) Quantum-Like Interdependence Theory Advances Autonomous Human–Machine Teams (A-HMTs). *Entropy* 22(11):1227. <https://doi.org/10.3390/e22111227>
- Lawless WF, Moskowitz IS, Doctor KZ (2023) A Quantum-like Model of Interdependence for Embodied Human–Machine Teams: Reviewing the Path to Autonomy Facing Complexity and Uncertainty. *Entropy* 25(9):1323. <https://doi.org/10.3390/e25091323>
- Li Q, Gkoumas D, Lioma C, Melucci M (2021a) Quantum-inspired multimodal fusion for video sentiment analysis. *Information Fusion* 65:58–71. <https://doi.org/10.1016/j.inffus.2020.08.006>
- Li Q, Gkoumas D, Sordani A, Nie JY, Melucci M (2021b) Quantum-inspired Neural Network for Conversational Emotion Recognition. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol 35, pp 13270–13278. <https://doi.org/10.1609/aaai.v35i15.17567>
- Li Q, Wang B, Melucci M (2019) CNM: An Interpretable Complex-valued Network for Matching. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Association for Computational Linguistics, Minneapolis, Minnesota, pp 4139–4148. <https://doi.org/10.18653/v1/N19-1420>
- Lo KI, Sadrzadeh M, Mansfield S (2025) Quantum-Like Contextuality in Large Language Models. *Proceedings of the Royal Society A*. arXiv:2412.16806
- Lorenz R, Pearson A, Meichanetzidis K, Kartsaklis D, Coecke B (2023) QNLP in Practice: Running Compositional Models of Meaning on a Quantum Computer. *Journal of Artificial Intelligence Research* 76:1305–1342. <https://doi.org/10.1613/jair.1.14329>

- Lukac M, Perkowski M (2007) Quantum Mechanical Model of Emotional Robot Behaviors. In: Proceedings of the 37th International Symposium on Multiple-Valued Logic (ISMVL 2007). IEEE, Oulu, Finland
- López-Incera A, Ried K, Müller T, Briegel HJ (2020) Development of Swarm Behavior in Artificial Learning Agents that Adapt to Different Foraging Environments. *PLoS ONE* 15(12):e0243628. <https://doi.org/10.1371/journal.pone.0243628>
- Maksimovic M, Maksymov IS (2025) Quantum-Cognitive Neural Networks: Assessing Confidence and Uncertainty with Human Decision-Making Simulations. *Big Data and Cognitive Computing* 9(1):12. <https://doi.org/10.3390/bdcc9010012>
- Mannone M, Chella A, Pilato G, Seidita V, Vella F, Gaglio S (2024) Modeling Robotic Thinking and Creativity: A Classic–Quantum Dialogue. *Mathematics* 12(5):642. <https://doi.org/10.3390/math12050642>
- Mannone M, Seidita V, Chella A (2022) Categories, Quantum Computing, and Swarm Robotics: A Case Study. *Mathematics* 10(3):372. <https://doi.org/10.3390/math10030372>
- Meghdadi A, Akbarzadeh-T. MR, Javidan K (2022) A quantum-like cognitive approach to modeling human biased selection behavior. *Scientific Reports* 12:22545. <https://doi.org/10.1038/s41598-022-13757-2>
- Melnikov AA, Makmal A, Briegel HJ (2018) Benchmarking Projective Simulation in Navigation Problems. *IEEE Access* 6:64639–64648. <https://doi.org/10.1109/ACCESS.2018.2876494>
- Moreira C (2017) Quantum Probabilistic Graphical Models for Cognition and Decision. PhD thesis, Instituto Superior Técnico, Universidade de Lisboa
- Moreira C, Wichert A (2014) Interference effects in quantum belief networks. *Applied Soft Computing* 25:64–85. <https://doi.org/10.1016/j.asoc.2014.09.008>
- Moreira C, Wichert A (2016) Quantum-Like Bayesian Networks for Modeling Decision Making. *Frontiers in Psychology* 7:11. <https://doi.org/10.3389/fpsyg.2016.00011>
- Moreira C, Fell L, Dehdashti S, Bruza P, Wichert A (2020b) Towards a Quantum-Like Cognitive Architecture for Decision-Making. *Behavioral and Brain Sciences* 43:e17. <https://doi.org/10.1017/S0140525X19001687>
- Moreira C, Tiwari P, Pandey HM, Bruza P, Wichert A (2020a) Quantum-like influence diagrams for decision-making. *Neural Networks* 132:190–210. <https://doi.org/10.1016/j.neunet.2020.07.009>
- Paparo GD, Dunjko V, Makmal A, Martin-Delgado MA, Briegel HJ (2014) Quantum Speedup for Active Learning Agents. *Physical Review X* 4(3):031002. <https://doi.org/10.1103/PhysRevX.4.031002>
- Pushparaj K, Ky G, Ayeni AJ, Alam S, Duong VN (2021) A quantum-inspired model for human-automation trust in air traffic controllers derived from functional Magnetic Resonance Imaging and correlated with behavioural indicators. *Journal of Air Transport Management* 97:102143. <https://doi.org/10.1016/j.jairtraman.2021.102143>
- Ried K, Müller T, Briegel HJ (2019) Modelling Collective Motion Based on the Principle of Agency: General Framework and the Case of Marching Locusts. *PLoS ONE* 14(2):e0212044. <https://doi.org/10.1371/journal.pone.0212044>
- Roeder L, Hoyte P, van der Meer J, Fell L, Johnston P, Kerr G, Bruza P (2023) A Quantum Model of Trust Calibration in Human–AI Interactions. *Entropy* 25(9):1362. <https://doi.org/10.3390/e25091362>
- She L, Han S, Liu X (2021) Application of quantum-like Bayesian network and belief entropy for interference effect in multi-attribute decision making problem. *Computers & Industrial Engineering* 157:107307. <https://doi.org/10.1016/j.cie.2021.107307>
- Sinha A, Macaluso A, Klusch M (2025) Nav-Q: Quantum Deep Reinforcement Learning for Collision-Free Navigation of Self-Driving Cars. *Quantum Machine Intelligence* 7(1):19. <https://doi.org/10.1007/s42484-024-00226-4>
- Skolik A, Jerbi S, Dunjko V (2022) Quantum Agents in the Gym: A Variational Quantum Algorithm for Deep Q-Learning. *Quantum* 6:720. <https://doi.org/10.22331/q-2022-05-24-720>
- Song Q, Wang W, Fu W, Sun Y, Wang D, Gao Z (2022) Research on Quantum Cognition in Autonomous Driving. *Scientific Reports* 12:300. <https://doi.org/10.1038/s41598-021-04239-y>
- Sordoni A, Nie JY, Bengio Y (2013) Modeling term dependencies with quantum language models for IR. In: Proceedings of the 36th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '13). ACM, Dublin, Ireland, pp 653–662. <https://doi.org/10.1145/2484028.2484098>
- Stoudenmire E, Schwab DJ (2016) Supervised Learning with Tensor Networks. In: Advances in Neural Information Processing Systems 29 (NIPS 2016). Curran Associates, Inc., pp 4799–4807. <https://proceedings.neurips.cc/paper/2016/hash/5314b9674c86e3f9d1ba25ef9bb32895-Abstract.html>
- Svozil K (2025) Quantum Contextuality for Contextual Word Embeddings. Preprint at arXiv:2504.13824
- Uprety S, Liu H, Jaiswal AK, Song D (2024) Investigating Context Effects in Similarity Judgements in Large Language Models. Preprint at arXiv:2408.10711

- van der Meer J, Hoyte P, Roeder L, Bruza P (2025) Modeling the quantum-like dynamics of human reliability ratings in Human–AI interactions by interaction dependent Hamiltonians. Preprint at arXiv:2504.13918. arXiv preprint
- Wei Q, Ma H, Chen C, Dong D (2022) Deep Reinforcement Learning with Quantum-Inspired Experience Replay. *IEEE Transactions on Cybernetics* 52(9):9326–9338. <https://doi.org/10.1109/TCYB.2021.3053414>
- Wichert A, Moreira C, Bruza P (2020) Balanced Quantum-Like Bayesian Networks. *Entropy* 22(2):170. <https://doi.org/10.3390/e22020170>
- Widdows D, Peters S (2003) Word Vectors and Quantum Logic: Experiments with negation and disjunction. In: *Proceedings of the 8th Mathematics of Language Conference (MOL 8)*, Bloomington, IN, pp 141–154
- Widdows D, Rani J, Pothos EM (2023) Quantum Circuit Components for Cognitive Decision-Making. *Entropy* 25(4):548. <https://doi.org/10.3390/e25040548>
- Yan F, Iliyasu AM, Jiao S, Yang H (2019) Quantum Structure for Modelling Emotion Space of Robots. *Applied Sciences* 9(16):3351. <https://doi.org/10.3390/app9163351>
- Yan F, Iliyasu AM, Hirota K (2021a) Conceptual Framework for Quantum Affective Computing and Its Use in Fusion of Multi-Robot Emotions. *Electronics* 10(2):100. <https://doi.org/10.3390/electronics10020100>
- Yan F, Yang X, Li N, Yu X, Zhai H (2021b) Emotion Generation and Transition of Companion Robots Based on Plutchik's Model and Quantum Circuit Schemes. *Security and Communication Networks* 2021:6802521. <https://doi.org/10.1155/2021/6802521>
- Yukalov VI (2023) Quantum operation of affective artificial intelligence. *Laser Physics* 33(6):065204. <https://doi.org/10.1088/1555-6611/accf7a>
- Yukalov VI, Yukalova EP, Sornette D (2018) Information Processing by Networks of Quantum Decision Makers. *Physica A: Statistical Mechanics and its Applications* 492:747–766. <https://doi.org/10.1016/j.physa.2017.11.004>
- Yukalov VI, Yukalova EP, Sornette D (2022) Role of collective information in networks of quantum operating agents. *Physica A: Statistical Mechanics and its Applications* 598:127365. <https://doi.org/10.1016/j.physa.2022.127365>
- Yun WJ, Kwak Y, Kim JP, Cho H, Jung S, Park J, Kim J (2022) Quantum Multi-Agent Reinforcement Learning via Variational Quantum Circuit Design. In: *Proceedings of the 2022 IEEE 42nd International Conference on Distributed Computing Systems (ICDCS)*. IEEE, pp 1332–1335. <https://doi.org/10.1109/ICDCS54860.2022.00151>
- Zhang C, Li Q, Song D, Tiwari P (2025) Quantum-Inspired Semantic Matching Based on Neural Networks with the Duality of Density Matrices. *Engineering Applications of Artificial Intelligence* 140:109667. <https://doi.org/10.1016/j.engappai.2024.109667>
- Zhang P, Su Z, Zhang L, Wang B, Song D (2018) A Quantum Many-body Wave Function Inspired Language Modeling Approach. In: *Proceedings of the 27th ACM International Conference on Information and Knowledge Management (CIKM 2018)*. ACM, pp 1303–1312. <https://doi.org/10.1145/3269206.3271723>